

4. Lichens can be used as indicators of the level of air pollution.

There are three types of lichen – bushy, leafy and crusty.
The photograph shows the three types of lichen.



The table below shows how the presence (✓) or absence (x) of each type of lichen in an area allows an assessment to be made of the level of air pollution.

Type of lichen	No pollution	Low pollution	Moderate pollution	High pollution
bushy	✓	x	x	x
leafy	✓	✓	x	x
crusty	✓	✓	✓	x

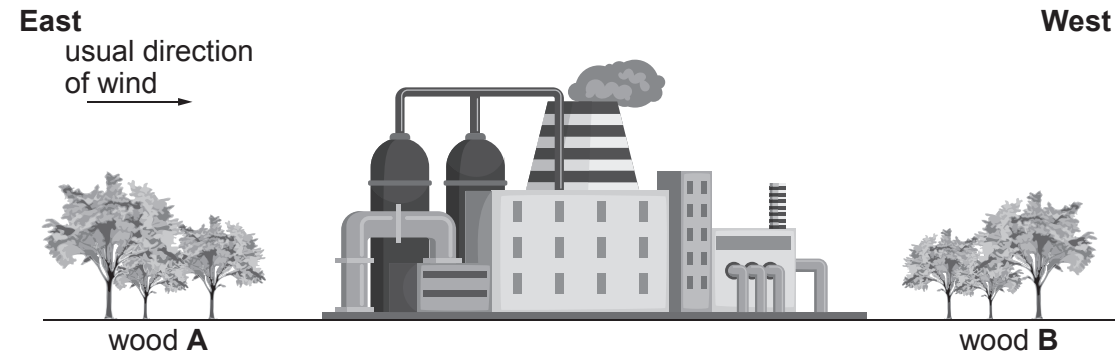
(a) Sharon examined several trees in her school grounds. She used the photograph above to identify each type of lichen she found. She found only leafy and crusty lichens.

State the level of air pollution indicated by Sharon's findings. [1]

.....



(b) The drawing shows two woods, **A** and **B**, separated by a factory.



Sharon investigated air pollution in the woods. She made the following hypothesis:
*'Air pollution is higher in wood **B** than in wood **A**'.*

(i) State the evidence in the drawing that Sharon used to make her hypothesis. [1]

(ii) Using lichens on trees as indicators, design an investigation to test the hypothesis:
*'Air pollution is higher in wood **B** than in wood **A**'.*
Explain how you would use the results to support **or** disprove the hypothesis.
[6 QER]

